Red-team of the Galbraith MPKC v2 intake (PR #2945) and of the book's literal statements

- **Date:** 2026-11-03
- **Kind:** adversarial evidence review. This is a review record, not a status change. It edits no PR #2945 file, catalog row or formal target.
- **Subject:** PR #2945 (codex/galbraith-mpkc-formalization-20261101, head 74a0d7f52eb2384f85008eed1e048dbe93539218), which added KN-LIT-2e5200, the 1,817-object catalog under inputs/GALBRAITH-MPKC-2018/ and the Lean target galbraith-montgomery-reduction-20261101. Also the theorem-like rows of the source book: Steven D. Galbraith, Mathematics of Public Key Cryptography v2.0, PDF SHA-256 ca75ee62b96daa0d46faaa2d4dc28205683455ad3a2a48586b9b39299c25022d (re-downloaded 2026-11-03; the hash matches), with the official errata page snapshot of 2026-11-03.
- **Question:** which of the proofs, lemmas, bounds and theorems that this intake brought into the program can be disproved, and which survive?

Figure (kept out of this PDF for the 64 KiB artifact limit): verdicts.svg / verdicts.png in this report directory.

1. Summary

Object	Verdict
Lean montgomery_reduction (Lemma 2.5.3)	**Survives.** I re-proved the exact proposition independently in core Lean 4.35.0 without Mathlib; its axioms are propext, Classical.choice and Quot.sound. Exhaustive and box searches found no counterexample. Three fidelity findings, all strengthenings (§2).
Theory note of the target	Two inaccuracies. The printed statement does not contain w < n; only the printed proof does. The inverse form needs only the negative-inverse hypothesis, not "odd n, R = 2^s" (§2).
Catalog: 1,817 objects, kind counts, no gaps or duplicates	**Confirmed**, by an independent detector (§3).
Catalog: SHA-256 reproduction	**Holds only under PyMuPDF 1.26.6**, which the README does not pin. PyMuPDF 2.1 gives a different hash and different class counts (§3).
Catalog: 119 absent_in_book proofs	**Refuted.** At least 11 of the 119 have proofs; the true count is at most 108. At least 10 citation-only proofs are classed present_complete (§3).
Catalog: 444 exercises with an Appendix B signal	**Refuted.** The count is 486; 42 are missed (§3).
Catalog: errata mapping	**Partly refuted.** ERRATA-041 (the Wiener bound) is attached to Lemma 2.3.3 instead of Theorem 24.5.2. There is edition-number drift for 5.2.20⟶5.2.25, 5.3.8⟶5.3.10 and "7.3.1"⟶17.3.1 (§3).
Book, Chapter 2 (the planned next formal tranche)	12 rows false as printed, 15 missing a hypothesis or ill-posed, 3 proof gaps, 4 constant/unit/typo defects, 32 survived. Lemmas 2.5.1 and 2.5.4, the named next targets, are **not well-formed Lean targets as printed** (§4).
Book, Chapters 13–15 (DLP and index-calculus bounds)	7 false as printed, 1 missing hypothesis, 3 proof gaps, 4 constant mismatches, about 40 survived. The core constants survived (§5).
Book, Theorem 24.5.2 (Wiener)	The printed bound N^{1/4}/√3 is a known erratum (√6). I found **941 concrete failures of the printed proof route** inside the printed bound (§6).

None of this shows that the book is unreliable in general. It shows that literal transcription into Lean targets or baseline constants would encode false statements, which is the risk the PR's own policy anticipates.

2. The Lean target and its source (verified by the lead)

Source (PDF p. 52, read from the pinned PDF). Lemma 2.5.3 assumes odd n, R = 2^s > n, n' = −n^{-1} mod R with 1 ≤ n' < R, a in Montgomery representation (0 ≤ a < n), and u = a n' mod R. It concludes w = (a+un)/R ∈ Z and w ≡ aR^{-1} (mod n). The bound w < n appears only in the printed proof.

Checks:

1. **Book statement, exhaustive** (evidence/lead/lemma253_check.py). All odd n < R = 2^s for s ≤ 10 and all a < n were tested: 349,525 cases, 0 failures. For every case with s ≤ 8, w is exactly the canonical representative of aR^{-1} mod n.
2. **PR proposition over a box.** All of n, R, n', a, u were taken from [−6, 12]. No counterexample was found.
3. **Mutation (drop one hypothesis).** Counterexamples exist when any of ha0, han, hu0, huR, hnegInv or hu is dropped, so each is load-bearing. Dropping **hnpos or hnR gives no counterexample**. Both are redundant: 0 ≤ a < n gives n > 0, 0 ≤ u < R gives R > 0, and R > n is never used by the integer core.
4. **Independent machine check** (lean/MontgomeryReductionCore.lean). It uses core Lean only, with grind and omega instead of Mathlib's ring/nlinarith, and compiles with lean v4.35.0. It proves:
   - montgomery_reduction_core: the PR's exact proposition;
   - montgomery_reduction_core_min: the same without hnpos and hnR;
   - montgomery_inverse_form: n*n'+1 = k*R and n | R*w − a imply n | w − a*k. The book's inverse form w ≡ aR^{-1} therefore follows from the negative-inverse hypothesis alone, contrary to the theory note, which attributes it to "odd n, power-of-two R";
   - a decided control: for Montgomery-multiplication inputs a < nR, the bound w < n fails (n=3, R=4, a=11, w=5). The book's own sketch says only w < 2n there, and that holds.

   The PR's Mathlib-based build was not reproduced here: the Mathlib cache does not fit in this session's disk allowance. The core-Lean re-proof is independent evidence for the same proposition.

Recommendation for the target: record the two redundant hypotheses and the inverse-form theorem as a dated correction or follow-up target. Do not edit the frozen spec. Fix the note's sentence about the source statement containing w < n.

3. Catalog claims (catalog audit agent; key rows spot-checked by the

lead) Scripts and results are in evidence/catalog-audit/. These were produced by a delegated audit agent. The lead re-read the PDF text for Theorems 24.2.14 and 24.2.16 ("Proof: (Theorem 24.2.14" and "Proof of Theorem 24.2.16" on p. 518) and for the ERRATA-041 text.

- **Reproduction.** With PyMuPDF 1.26.6, the catalog SHA-256 764d1249…789d, the summary, the errata JSON and every index are byte-identical. With PyMuPDF 2.1, the object count is still 1,817 but the hash is 0e473889…e752. That run also gives absent 122, citation 52, present_complete 257 and Appendix B 440, and 381 index lines differ. The generator is not reproducible without a pinned PyMuPDF version, and the README does not pin it.
- **Completeness.** An independent bold-heading detector found 1,806 labelled headings, plus 9 Notation blocks and 2 unnumbered objects (p. 93, p. 292), for 1,817 in total. It found 0 spurious, 0 missed, no numbering gaps or duplicates, and every kind count matches.
- **Proof status.**
  - A random sample (seed 20261103) found 2 of 25 absent_in_book rows wrong.
  - A full scan of all 119 found 11 wrong:
    - 8 have the proof in the statement's own PDF block: 2.9.1, 9.6.13, 10.1.14, 14.1.1, 19.3.1, 22.1.5, 24.2.16, and 10.7.4 (really citation-only).
    - 3 have the proof displaced after another object: 7.7.11, 11.1.4 and 24.2.14.
  - At least 10 rows whose proof is only a literature pointer are classed present_complete: 8.7.2, 9.10.12, 9.11.2, 10.2.6, 10.2.11, 10.9.5, 19.2.1, A.6.7, A.7.2 and A.10.8.
  - The problem record's "119 theorem-like rows have no detected in-book proof" therefore overstates the queue by about 9%.
- **Appendix B.** An independent scan of pp. 605–642 found 486 exercise entries. All 444 catalog rows are among them; 42 are missed (for example 2.15.1, 10.2.4, 13.3.3 and 24.6.3) because get_text(sort=True) merges lines.
- **Errata.**
  - The 47 items are confirmed.
  - The 31 exact matches cover 31 objects but 33 errata ids.

- **ERRATA-041** ("correct bound … $0 < d < N^{1/4}/\sqrt{6}$") is attached to Lemma 2.3.3, which it only cites. It is missing from v2 **Theorem 24.5.2** (p. 528), which still prints $\sqrt{3}$.
- 2012-to-v2 drift: Lemma 5.2.20 is v2 Lemma 5.2.25, Theorem 5.3.8 is v2 Theorem 5.3.10, and the erratum's "Definition 7.3.1" means v2 Definition 17.3.1.
- No erratum touches §2.5 or Montgomery reduction, as the PR claims.

4. Chapter 2 literal statements (Chapter 2 agent; spot-checked by the

lead) Scripts and outputs are in evidence/ch2/. The scripts read page text they extract from the pinned PDF; the book text itself is not committed. The lead re-read three rows against the PDF:

- Lemma 2.3.3(1): the book's own Example 2.3.4 has $s_1 = 1, s_2 = -1$, which contradicts $|s_i| < |s_{i+1}|$ for $i \geq 1$.
- Lemma 2.14.8: the statement omits $\mathrm{Tr}(\alpha) = 0$, which appears only in the preceding Exercise 2.14.7.
- Lemma 2.5.4: as printed, it carries no bound on s.

Row	Defect	Evidence
Lemma 2.5.1(4)	Ill-posed: the bound $O(\log n\,(\log a - \log n + 1))$ is negative for $a < n/e$ and undefined for $a \leq 0$. The book's log is natural (p. 15) and Def. 2.1.2 requires $g > 0$.	$(a,n) = (1,100)$ gives -16.6
Lemma 2.5.1(1–3)	The bound is 0 at $n = 1$. Item 3 needs $\gcd(a,n) = 1$.	$a = 2, n = 6$
Lemma 2.5.4	Missing $s = O(\log n)$: nothing bounds $R = 2^s$, so the cost is $O(M(s))$. It is also unstated whether the cost of n' is charged.	argument
p. 53, Newton/Barrett text	a and n are swapped. Even after the fix, $\lfloor ab/2^e \rfloor \neq \lfloor a/n \rfloor$ in 2,418,140 of 21,253,395 cases (off by 1), so a correction step is needed.	t1.out
p. 53, CRT formula	The result is not reduced into $[0, m_1m_2)$: 87% of cases are out of range.	$(2,3,1,0)$ gives -3
Algorithm 1 (symmetric remainders)	Returns a negative "gcd".	$(2,3)$ gives $d = -1$; 1,442 pairs
Lemma 2.3.3(1), (6)	False at $i = 1$, including on the book's own example.	Example 2.3.4
Lemma 2.3.3(7), (8)	Missing $\gcd(s,t) = 1$, and missing $s, r \neq 0$.	$(1,1,2,-2)$
Thm 2.3.1 (proof)	$2\lvert r_i\rvert \leq \lvert r_{i-1}\rvert$ does not imply $1 + \ln r_i \leq \ln r_{i-1}$.	$(12,5,2)$
Lemma 2.9.2, Ex. 2.9.10	Missing p odd, $p \nmid a$, and $(a/p) = 1$.	$p = 3, a = 2$
p. 60, Cornacchia (2p variant)	Misses 2,078 of 2,918 solvable cases.	$7 = 2^2 + 3\cdot 1^2$
p. 57, Stolarsky	States a lower-bound conjecture as an upper bound.	$m = 127$ needs $10 > 9.80$
Thm 2.14.2	$q - m(m-1)$ should read "at least".	$q = 7, m = 3$
Lemma 2.14.8 (half trace)	Missing $\mathrm{Tr}(\alpha) = 0$: fails on exactly the trace-1 half.	$m = 3, 5, 7$
p. 69, order	"smallest f" should be "largest f".	$f = 0$ always qualifies
Lemma 2.15.7	The count fails with natural log, and $m = 1$ is undefined.	$m = 2$ gives $35 > 27.7$
p. 71, φ bound	Fails at $q = 3$.	$0.5 < 1.77$

Ex. 2.6.4	The claims $2z \notin Z$ and that x exists both fail.	all $c_i = 0$

The full table, with all survivors, is in evidence/ch2/ (t*.out). The items here are agent-computed except the three rows above.

Consequence for the campaign. The problem record's next step is to formalize "Lemmas 2.5.1 and 2.5.4". Neither can be transcribed faithfully: each needs a declared cost model first (bit lengths instead of logs, a rule for the case $a < n$, a bound on R, and which precomputation is charged). That is the "unmodelled complexity convention" stop condition the record already names.

5. DLP and index-calculus bounds, Chapters 13–15 (DLP agent; one row

re-derived by the lead) Scripts and outputs are in evidence/dlp/. "C" means exact computation or seeded Monte Carlo with 95% intervals; "A" means argument only.

Row	Verdict	Evidence
Thm 15.8.2 (summation polynomials)	**False over F_q as printed.** It holds over the algebraic closure; F_q-roots can come from the quadratic twist.	C. The lead re-derived this independently (evidence/lead/semaev_s3_fp.py): on $y^2 = x^3+2x+3/F_{31}$, 462 of 930 F_p-roots of S_3 come from no $E(F_p)$ points; on $y^2 = x^3+x+5/F_{37}$, 666 of 1332. Every point triple is a root.
Thm 14.5.5 and Pollard kangaroo data	The book quotes $C(n_S)$ between 1 and 1.06 for $n_S \geq 12$. The book's own recursions give $C(12) = 1.118$ and $C(13) = 1.110$ (simulation 1.115 ± 0.011). A full kangaroo with 2^j jumps gives 2.107 ± 0.046 ($n_S = 10$) and 2.140 ± 0.046 ($n_S = 12$). The asymptotic $2 + o(1)$ is not refuted.	C, t3_kangaroo.py
Ex. 14.4.8 (Koblitz)	$\sqrt{(\pi 2^m/(4m))}$ uses 2^m instead of $r = 2^m/h$, which overstates by $\sqrt{h}$.	A
p. 297, Brent–Pollard	$c = n_S/(n_S-1)$ is not the in-degree variance; the variance is the inverse. The constant fails for small n_S ($n_S = 3$: 2.04 against 1.22).	C
Lemma 13.5.6	The lower bound $\Omega(\#S_1^{l/2})$ needs $\Pi \#S_j \approx r$.	A
p. 286, Wiener birthday bound	Fails in the printed range $l \leq 2N \log N$ (first at $l = 181$, $N = 1000$).	C
p. 287, Kuhn–Struik	$\sqrt{(2kN)}$ is 6.4% high at $k = 2$.	C
Thm (p. 292) Floyd failure probability, proof	Assumes $b_{i+lh} - b_i$ is constant, which is false for the squaring walk (190 of 197 cycles). The 3.093 constant survives.	C
Lemma 14.6.5 (proof)	The colliding-pair formula has a sign error (3,689 of 4,000 cases).	C
Alg. 14 (BSGS)	Claimed to solve the decision problem, but has no loop bound, so it does not halt when $h \notin \langle g \rangle$.	C

Thm 13.6.9	"WLOG n, w even" breaks Algorithm 15.	C
§15.5.1, Ex. 15.5.7	"Rigorous" L(1/2, 2) omits the full-rank argument.	A
Conj. 15.5.21	The optimisation text and the numeric value: 2^68.9, not 2^67. Partly KNOWN.	C
§15.8.2 Gaudry degree	The eliminant degree is n!·2^{n(n−1)} in the x-variables.	C
Lemma 15.1.6(8)	= should be ≤.	A, F ≡ 1

Survived (C): - Thm 14.1.1; - birthday examples; - epact constant 0.8226; - Floyd 3.0929; - tail/cycle √(πN/8); - **rho √(πr/2)** - **Gaudry–Schost 2.43284** (exact lattice sum); - BSGS and Algorithm 13; - Lemmas 13.4.4, 13.6.6 and 13.6.7; - the Certicom ratios; - the stated L-values.

Survived (A): the negation-map √(πr/4) heuristic, the three- and four-kangaroo constants, and Theorems 13.2.5, 13.3.1, 14.2.16, 14.3.2, 14.6.2, 14.6.6, 14.9.2, 15.2.7 and 15.3.2. Cited external results (CEP, Diem, Enge–Stein and others) were not re-proved.

Leaderboard relevance (interpretation). - The matched-rho references the program uses (√(πr/2), and √(πr/4) with negation) survived. - Any kangaroo reference built on "C(n_S) ≤ 1.06" or a flat 2√w at practical n_S is low by about 5–7%. - Any Koblitz reference computed from Ex. 14.4.8 is high by √h. - Any summation-polynomial decomposition test that accepts F_q-roots of Summ_n as relations without lifting the y coordinates over F_q over-counts by about half for S_3 in these instances.

I did not audit whether any existing row uses these formulas. That is the next check.

6. Theorem 24.5.2 (Wiener): a concrete failure of the printed proof

(verified by the lead) v2 prints 0 < d < N^{1/4}/√3; the official erratum (ERRATA-041) corrects this to √6. evidence/lead/wiener_window.py takes all p < q < 2p with p, q < 2000 and every d in the window [N^{1/4}/√6, N^{1/4}/√3) with gcd(d, φ) = 1.

- In **941 of 34,211** cases, d is not an e-cofactor of the Euclidean algorithm on (e, N), which is the route the printed proof uses.
- Smallest example: N = 137·167 = 22879, e = 19351, d = 7 < 7.10. The XGCD cofactors are 1, 1, 6, 13, 214 and 1511.
- Positive control: below the corrected √6 bound there are **0 failures in 18,597** cases.

This refutes the printed proof, not the existence claim. Polynomial-time extensions beyond N^{1/4} exist in the literature (recalled, not verified here). The catalog should carry this erratum on 24.5.2.

7. Recommended follow-ups (none executed here)

1. **PR #2945 catalog:**
   - pin pymupdf==1.26.6 in the README and record it in provenance.json;
   - extend the proof classifier to proofs in the same block and displaced proofs (11 rows), and to "Proof: see/This is ..." citations (at least 10 rows);
   - parse Appendix B without sort=True (42 rows);
   - move ERRATA-041 to Theorem 24.5.2 and record the three edition-drift maps.
2. **Formal lane:**
   - file a correction or follow-up target for the minimal Montgomery statement and the inverse form;
   - before any Chapter 2 complexity target, write a cost-model note. The Chapter 2 table above is a ready-made counterexample corpus: each row is a formal_counterexample candidate.
3. **Baselines:** audit leaderboard and ecbench references for the kangaroo C(n_S), the Koblitz √h factor, and summation-polynomial relation acceptance over F_q (Thm 15.8.2).

Provenance and limits

- **Verified by the lead in this session** (code in evidence/lead/ and lean/):
  - Lemma 2.5.3 and the Lean statement;
  - Thm 15.8.2, re-derived independently;

  - the Wiener window;
  - three Chapter 2 rows and two catalog rows, re-read against the PDF.
- **Computed by delegated agents** (Sonnet: catalog; Opus: Chapter 2, Chapters 13–15), with their scripts and outputs committed unchanged. These are not independently re-run unless listed above.
- **Not covered:**
  - Chapters 1, 3–12 and 16–26, apart from Thm 24.5.2;
  - unlabelled prose beyond the rows listed;
  - re-proving cited external theorems;
  - a Mathlib build of the PR file.
- **Copyright:** no book text, page images or the full catalog JSONL are committed. Scripts regenerate what they need from the pinned PDF URL and hash.